SOMAIYA VIDYAVIHAR

Somaiya TRUST

K J Somaiya Institute of Engineering and Information Technology
An Autonomous Institute permanently affiliated to University of Mumbai.

Department of IT

Cover page ASSIGNMENT Academic Year: 2026-27 Class: SY Sem: III

Subject: DSA Course Code:

Name: Waghela Jinal Nitin Roll No: 52-B/BS3

ASSIGNMENT NO : 1/2 Assignment-I		Date of Performance: 23.10.2026	Date of Correction: 23.10.2026
Rubrics	**Marks (20)**	**Grade**	**Signature of Faculty**
Participation and Involvement (5)	5	18 E	Jinal
Preparation and Knowledge (5)	4		
Accuracy and neatness of Documentation(5)	5		
Punctuality (5)	4		

Grades:
Excellent – E (17 – 20)
Very Good – A (13-16)
Average – B (9-12)
Satisfactory – C (5-8)
Unsatisfactory – D (< - 5)

Assignment - I

⇒ Code:-

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#define max 5.

struct Observation {
        int id;
        char location[30];
        char timestamp[15];
        float temp;
        float humidity;
    };
struct Observation q[max];
int front=-1; real=-1; search-id;
found, countval;

Insertion
void insertion()
    { if real == max-1) {
            printf("\n OVERFLOW \n");
    }
    else {
        if (real==-1 && front==-1) {
            real=front=0;
    }
    else {
            real++; }
    printf (" Enter ID");
```

```
printf("Enter timestamp;");
printf("%f", &q[rear].temp);
printf("Enter humidity");
scanf("%f", &q[rear].humidity);

display();
}}
```

2] Deletion

```
void deletion(){
    if (front==-1 || front>rear){
printf("UNDERFLOW");
}
else
{
printf("Deleted Observation");
printf("ID: %d ... location: %s .. Temp %f*(
printf("humidity ...)
  q[front].location, q[front].temp;
 q[front].humidity);
front++;

if (front>rear)
  {
   front=rear=-1;
}}}
```

3] Peek

```
void peek(){
  if(front==-1 || front > rear){
printf("QUEUE is EMPTY");
}
else
printf("ID: %d ... Location: %s -- Temp: %f *
  C -- Humidity: %f %if)",
q[front].id, q[front].location, q[front].temp
, q[front].humidity;
}}
```

4] Display

```
void display(){
  if (front==-1 || front>rear)
printf("Queue is empty");
printf("\n");
} else{
printf("Queue element")
for(i=front; i<=rear; i++)
}}
```

5] Count

```
void count(){
  if(front==-1 || front > rear){
printf("Total observation=0");
}
else{
countval = (rear-front)+1;
printf("Total Observation ... %d\n", count
}}
```

Department of: - /Class /Sem. /Sub. /Academic Year

7] Queue Status

```
void queuestatus (){
  if (front == -1 || front > rear) {
  printf ("Empty (underflow)");
  }
  else {
  printf ("front = %d, Real = %d \n", front, rear);
  }
int main(){
   int c;
printf ("ARRAY SIZE = %d \n", max);
do{
 printf ("1. Insertion, 2. Deletion, 3. feat
        4. Display \n");
printf ("5. Search, 6. count, 7. status, 8. exit \n");
 printf ("Enter choice:");
  scanf ("%d", &c);
switch (c){
case1: printf ("Insertion");
       insertion();
        break;
case2: printf ("Deletion");
       deletion();
       break;
case3: peek();
       break;
case 4: display();
       break;
case 5: search();
       break;
```

```
case 6: count();
        break;
case 7: queuestatus();
        break;
case 8: exit(0);
        break;
default;
printf("Invalid choice");
} } while (c!=8);
return 0;
}
```

Time & Complexity

insert()	O(1)	Insert at rear
del()	O(1)	Removes from front
peek()	O(1)	Access front element
display()	O(n)	Works every time
search()	O(n)	May examine all elements
count()	O(1)	Use count variable
overstatus()	O(1)	checks count & capacity.

Space complexity

O(M), where M is fixed queue capacity. To array allocates space for a/c observation regardless of how many are currently stored.

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